

Real-World Results & Validation

An evidence-first account of one traceable video-to-3D run: what completed, what was measured, and what remains unverified.

Research, engineering and validation evidence

EXECUTIVE VIEW / 03

What AeroForge3D has demonstrated so far

Evidence cut-off: 29 September 2026 run artifacts; project status cross-checked against the README dated 30 September 2026.

AeroForge3D prepared a 193.6-second drone video with 194 deterministic one-second frames, matched all prepared frames to valid SRT records, and completed one full georeferenced reconstruction after recovery from three failed attempts. The integrated workspace rendered a textured mesh and point cloud; the orthophoto, DSM, and primary downloads were verified; and distance, area, and DSM-volume records survived reload and GeoJSON export.

The result is a working end-to-end prototype on one dataset, not a validated survey or automated inspection product. The reconstruction is visibly fragmented. No independent checkpoints or known reference dimensions were used. The run's reported reprojection and GPS statistics are software-generated diagnostics, and the altitude units/datum are unknown. Low-light use, defect detection, Gaussian Splats, and the under-10-minute target remain unvalidated or planned.

Evidence in one sentence: the workflow now reaches usable files and persistent review records on a real capture, while coverage completeness and real-world measurement accuracy remain open validation gates.

Reader's map

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01 / EXPERIMENTAL SETUP

One real capture, with source and processing provenance

The validation unit is one flight video and its matched telemetry. The run is not a multi-site benchmark.

Item	Recorded setup
Drone/video	DJI-tagged H.265/HEVC MP4; 5472 x 3078; 30000/1001 fps; 193.5934 s; 5,802 reported frames; full decode exit 0. Exact aircraft/camera model is not established in the run record.
Scene	The reviewed frame depicts a dry, open scrubland site with a central dirt road. No independent site survey or known dimensions accompanied the capture.
Telemetry	DJI SRT, 5,801 records parsed; all valid with no parser errors, duplicates, non-monotonic sequence, or gaps greater than one second. Altitude values range 21.6-29.7; units and vertical datum are unknown.
Frame preparation	Frozen deterministic R1: first decoded frame at/after each 1 s interval boundary; FFmpeg seek extraction; 194 ordered 5472 x 3078 JPEGs with hashes and manifest.
Geolocation	All 194 images matched to SRT records within 0.5 s; EPSG:4326 inputs. Median timestamp difference 0.00558 s; P95 0.01017 s; maximum 0.01117 s.
Compute/software	Containerized Linux processing with memory capped at 7.607 GiB; concurrency set to 1. Two attempts failed under memory pressure. Host CPU/GPU model and exact container image digest were not recorded in the run evidence.
Reconstruction options	GLB, DSM, and map tiles enabled. Fourth attempt resumed from a later stage and used the full 3D mesh path for orthophoto generation; a separate 2.5D textured surface was omitted.
Reference truth	No surveyed checkpoint set, independent surface, known reference length, or verified vertical datum. Metric accuracy cannot be inferred from this run.

Observed end-to-end path

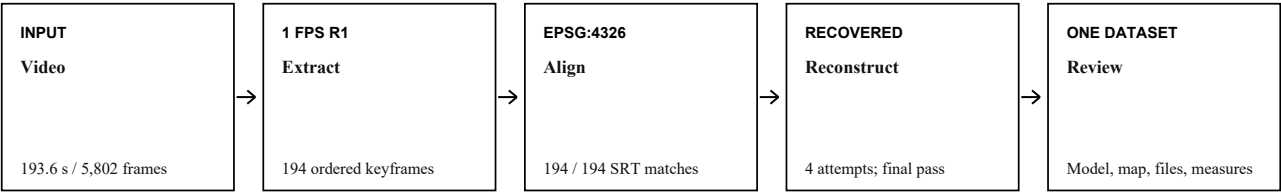


Figure 1. Actual workflow for the 29 September run. Frame selectors beyond the fixed R1 baseline are not shown because they were not used for this reconstruction.

02 / QUANTITATIVE RESULTS

One run produced useful diagnostics - and unresolved quality

Values below are direct run artifacts. Processing-report estimates are explicitly separated from independent validation.

194 frames retained From 5,802 reported source frames; deterministic 1 fps.	194 / 194 images registered 100% as reported by the reconstruction quality report.	2.16M dense points PDAL-read LAZ; projected CRS EPSG:32637.
0.93 px mean reprojection Software report diagnostic; not positional accuracy.	5 cm/px orthophoto spacing Raster metadata check; not a ground-truth accuracy result.	4 attempts to completion Three failures, then a resumed pass with an output-path change.

Table 1. Selected observed run counts and tool-reported diagnostics. Reprojection and raster spacing are different quantities.

Processing and output report

Measure	Observed value	Evidence class / limitation
Registered images	194 of 194 (100%)	Reconstruction quality report; does not measure spatial completeness.
Sparse points	115,346 of 180,042 (64.1%)	Software report count; denominator is report-defined.
Dense points	2,155,428	PDAL read from the georeferenced LAZ output.
Mean reprojection error	0.93 px	Image-space diagnostic; not a survey accuracy statistic.
Reported GPS errors / RMS	1.74 m / 1.737 m	Software-reported fit to image position inputs; not checked against independent checkpoints.
Report accuracy estimates	Horizontal CE90 0.757 m; vertical LE90 3.469 m	Estimated by processing report; no control/check points. Do not present as verified accuracy.
GSD values	Report prints 0.51 cm; orthophoto raster is approx. 5 cm/pixel	Inconsistent report/raster values are preserved; neither is used to claim accuracy.
Processing time	922.794 s cumulative task time across recorded attempts; final resumed segment 182.084 s	Not a benchmark. Input was 193.6 s, not a 10-minute benchmark clip; earlier failed work and a recovery option change are included.

Runtime target: under 10 minutes for a 10-minute video is an engineering target only. This run does not validate that target. It used a shorter clip and accumulated about 15 minutes 23 seconds across attempts.

03 / VISUAL EVIDENCE

The outputs are real; the missing coverage is visible

Images below are from the 29 September run. The orthophoto preview was downsampled for page layout; blank regions are output NoData / coverage gaps.

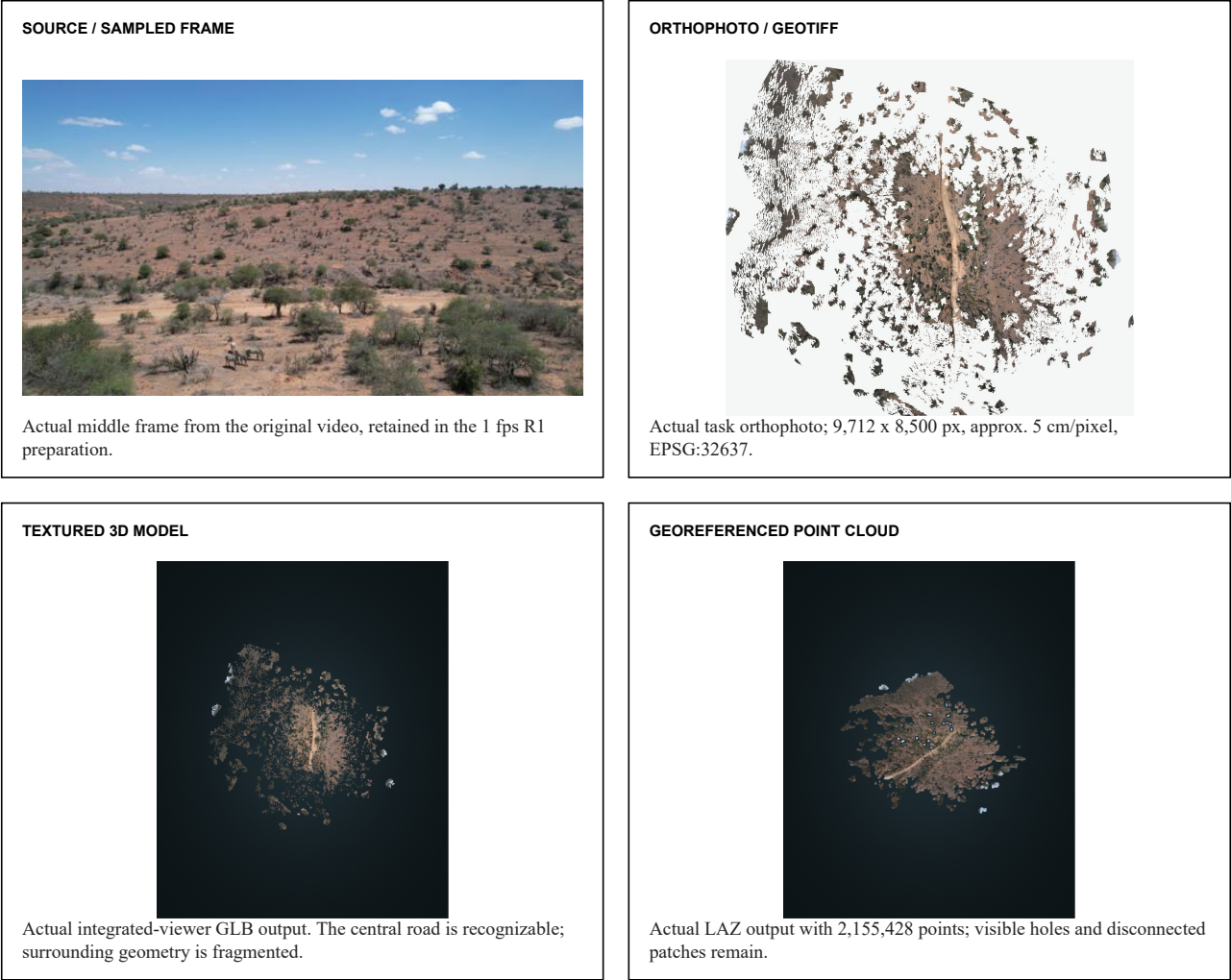


Figure 2. Actual project evidence: source frame, orthophoto raster, textured mesh view, and point-cloud view. The reconstructed footprint is incomplete; these images are not survey-quality certification.

Output	Validation performed
Textured GLB	Parsed successfully; 1 mesh, 62 primitives, 62 materials/textures; rendered in the integrated viewer.
LAZ point cloud	2,155,428 points read; CRS EPSG:32637; rendered in the point-cloud viewer.
Orthophoto GeoTIFF	9,712 x 8,500 pixels; four bands; approx. 5 cm/pixel; rendered as map tiles in workspace.
DSM GeoTIFF	9,711 x 8,499 pixels; approx. 5 cm/pixel; 63,098,962 valid cells and 19,434,827 masked cells; vertical datum unknown.
UI delivery	GLB, LAZ, DSM, and orthophoto downloads matched authenticated task files byte-for-byte by SHA-256.

04 / MEASUREMENT AND INTERFACE

Persistence passed; physical correctness remains open

The test verifies that named records survive the workspace flow. It does not compare these sample placements with independent ground measurements.

Saved record	Value shown	What was tested	Qualification
Demo - Distance	160.8174 m	LineString record saved, map reloaded, exported, reopened.	Two-point line along visible road; no survey tape or checkpoint reference.
Demo - Area	3,974.9729 m2	Polygon record saved, map reloaded, exported, reopened.	Interior demo polygon; area not checked against a surveyed boundary.
Demo - Volume	622.6369 m3	DSM volume record saved and reloaded; export contains a Polygon feature.	5,840.02 m2 footprint and triangulated base; vertical metre assumed; datum unknown; vegetation and patchy DSM included.

Table 2. Persisted demonstration measurements. Values are app output records and are illustrative, not independent ground truth.

Interface checks completed

Check	Result
Model view	Textured GLB rendered in the task workspace.
Point-cloud view	Georeferenced LAZ rendered in the 3D workspace.
Map view	Orthophoto tiles loaded; central road visible; incomplete coverage observed.
Persistent measures	Distance, area, and DSM-volume records all remained after reload.
GeoJSON export	Three features reopened successfully with EPSG:4326 geometry and the saved values.
Downloads	Four primary spatial outputs matched task API copies by size and SHA-256.
Focused test suite	34 focused application tests passed; 1 explicit opt-in live companion test skipped; application system check clean.
Viewer height tool	12.79 m readout restored in its 3D scene after reload, but is not a durable measurement type in the application API.

Interpretation. Persistence and file integrity are validated application behaviors. Absolute position, physical lengths, area, volume, vertical units, and defect measurement are not independently validated.

05 / VALIDATION STATUS

Implemented, demonstrated, experimental, and planned

A completed task is one evidence point. It does not imply general reliability across sites, sensors, or conditions.

Status	Capability	Evidence / boundary
Validated	Video preflight and metadata inspection; repeatable 1 fps extraction and manifest; SRT parsing and all 194 timestamp associations; persistent distance / area / volume records; file downloads and GeoJSON parsing.	Current run files, input validation, measurement review, asset validation, focused 34-test suite. These validate software behavior on this dataset.
Partially validated	End-to-end georeferenced reconstruction, model/map review, DSM-derived volume, recovery/resume.	One dataset completed after three failed attempts. Visual coverage is fragmented; DSM/height semantics lack an independently verified vertical datum.
Experimental	Frame selectors beyond the deterministic R1 baseline; LightGlue-assisted scoring; coverage-aware ranking.	Separate research branches exist. R1 remains the default. No controlled equal-input quality or runtime improvement has been established.
In development	Fresh integrated workflow hardening, selection evaluation, resource-aware recovery, output review.	Current run needed three failures and an output-path change. More site and resource cases are needed.
Planned	Gaussian Splat output; automatic crack/anomaly detection; structural diagnosis.	Not in current outputs. Each requires its own technical evaluation and careful product boundary.
Not validated / target	Low-light or night use; sub-meter absolute accuracy; under-10-minute processing target; survey-grade measurements.	No controlled low-light dataset, independent checkpoints, known reference dimensions, or benchmark protocol result was found.

Why the single run still matters

The run closes several integration questions: source video and telemetry can be prepared with provenance; the 3D/map workspace can render files; outputs can be downloaded consistently; and three measurement types can be saved and re-opened. It also turns engineering risk into visible next work: memory failures, a texturing crash, output gaps, and unverified metric truth are recorded rather than hidden.

Evaluator reading: AeroForge3D has a real, traceable prototype and a useful validation path. Its strongest next claim should be about dependable workflow behavior; survey accuracy and automated inspection need more evidence first.

06 / LIMITATIONS AND NEXT VALIDATION

Close the quality gap with references that are independent of the pipeline

These next steps improve the evidence base. They are not claims about completed work.

Open limitation	Next validation action	Pass evidence to retain
No independent spatial reference	Collect surveyed checkpoints distributed across plan and height; keep check points separate from adjustment; declare horizontal CRS and vertical datum.	Checkpoint RMSE and residual distribution with sample count, survey uncertainty, CRS, datum, and process settings.
Fragmented mesh, point cloud, orthophoto	Run repeated flights with strong overlap and varied views; score missing regions against an independently reviewed visible-surface mask.	Completeness by region, failure count, source-view visibility, and paired baseline comparison.
Memory and retry dependence	Record host CPU/GPU, worker image digest, RAM/VRAM, disk, concurrency, and inclusive wall time; cold-repeat the same job.	All failures and retries retained; median and range; no output or quality changes hidden by recovery.
No selector performance proof	Compare R1 and experimental selectors on identical candidate pools, cameras, settings, and independent sites.	Frame count, registration, spatial completeness, checkpoint error, and resource cost in a paired protocol.
Low-light not tested	Capture daylight, shadow, twilight, and controlled low-light clips with exposure/blur metadata.	Image quality, registration, surface completeness, and checkpoint geometry across lighting strata.
Measurement ground truth absent	Measure known baselines, bounded areas, and volumes with independent instruments; document surface class and vertical reference.	Absolute and relative error with units, method, uncertainty, and applicability limits.
Defect tools not implemented	Build an annotated inspection set and expert-reviewed workflow before adding detection or diagnosis claims.	Per-class precision/recall, localization/width error, failure modes, and qualified engineering review.

Evidence index

ID	Project evidence
E1	Project README and production workflow notes README.md; docs/PRODUCTION_BASELINE.md
E2	Run walkthrough and full input validation runs/DJI_0142-R1-demo-20260929-193104/DEMO_WALKTHROUGH.md; input-validation.md
E3	Machine-readable task metrics and asset checks runs/DJI_0142-R1-demo-20260929-193104/report-metrics.json; asset-validation.json
E4	Visual review and failure/retry analysis runs/DJI_0142-R1-demo-20260929-193104/quality-review.md; texrecon-failure-analysis.md; attempts/
E5	Measurement persistence, tests, and export checks runs/DJI_0142-R1-demo-20260929-193104/measurement-review.json; ui-download-verification.json
E6	Preparation, SRT association, and geolocation records runs/DJI_0142-R1-demo-20260929-193104/preparation_report.json; telemetry/telemetry_report.json; geolocation_report.json
E7	Current UI screenshots and source frames runs/DJI_0142-R1-demo-20260929-193104/screenshots/; frame-review/

Final conclusion. AeroForge3D has demonstrated a genuine video-to-3D/map workflow on one drone capture and verified that key outputs and task measurements persist through its review interface. Its present model is incomplete, its processing report is not independent accuracy evidence, and its planned visualization and automated inspection functions are not validated. This is a credible prototype result with a clear next validation program.